

```
In [1]: import pandas as pd
from shopsense.data_generator import generate_customers, generate_transactions
from shopsense.eda import compute_univariate_stats, churn_distribution_summary,

# Generate Data
customers_df = generate_customers(10000, random_state=42)
transactions_df = generate_transactions(customers_df, random_state=42)

print("--- Univariate Stats ---")
display(compute_univariate_stats(customers_df, ['age']))

print("\n--- Churn Summary ---")
display(churn_distribution_summary(customers_df))

print("\n--- Cohort Retention ---")
display(compute_cohort_retention(customers_df, transactions_df).head())

print("\n--- Outlier Detection ---")
display(detect_outliers_iqr(transactions_df, 'unit_price'))
```

--- Univariate Stats ---

	mean	median	std	skewness	kurtosis	missing_pct	unique_count
age	34.6665	34.0	9.630377	0.258089	-0.328872	0.0	52

--- Churn Summary ---

```
{'total_customers': 10000,
 'churned_count': 2512,
 'churn_rate': 0.2512,
 'churn_by_channel': {'email': 0.24276606179499755,
 'organic': 0.2525406504065041,
 'paid_search': 0.23869346733668342,
 'referral': 0.2571428571428571,
 'social_media': 0.26507856056766343},
 'churn_by_gender': {'F': 0.25797161077967495,
 'M': 0.24292353189691593,
 'Other': 0.26666666666666666},
 'churn_by_premium': {False: 0.25475237618809404, True: 0.2370259481037924},
 'mean_age_churned': 34.488853503184714,
 'mean_age_retained': 34.726095085470085}
```

--- Cohort Retention ---

	period	0	1	2	3	4	5	6	7
cohort_month									
2021-01	1.0	0.226415	0.283019	0.377358	0.339623	0.245283	0.245283	0.226415	
2021-02	1.0	0.274194	0.225806	0.258065	0.282258	0.225806	0.266129	0.241935	
2021-03	1.0	0.282828	0.262626	0.257576	0.313131	0.242424	0.267677	0.454545	
2021-04	1.0	0.270270	0.281853	0.262548	0.270270	0.262548	0.409266	0.401544	
2021-05	1.0	0.254717	0.254717	0.273585	0.257862	0.455975	0.399371	0.433962	

5 rows × 36 columns

```
{'q1': 306.86,  
  'q3': 1273.295,  
  'iqr': 966.4350000000001,  
  'lower_bound': -1142.7925,  
  'upper_bound': 2722.9475,  
  'outlier_count': 16169,  
  'outlier_pct': 0.135}
```

In []: